

AthenaK MHD-PIC Test Problem Description and Validation Atlas (Internal Draft)

ATHENAK COLLABORATION¹

¹*Internal Methods Note*

ABSTRACT

This document is a method-facing atlas of the current **AthenaK** test-problem suite for MHD, stand-alone PIC, and coupled MHD-PIC workflows. We organize the material as one task per test, then describe each task in a common four-part structure: what it tests, what the present result is, what that result implies, and what upgrade is still worth doing.

The figure set mixes quantitative diagnostics (convergence curves, growth-mode histories, anisotropy slopes, and drift traces) with state visualizations (particle-support maps and shock-structure panels). The core message is practical: the suite already establishes broad implementation credibility, but the highest-impact CR-shock storyline and several instability windows still require targeted redesign before final publication claims.

1. SCOPE AND READING STRATEGY

The purpose of this paper is straightforward: make the test suite legible enough that a reader can move from code-level checks to physics-level confidence without guessing what each gate is actually proving. We therefore keep the mapping explicit. Every task description includes:

1. the *logic* of the test,
2. the present *result*,
3. the practical *implication*, and
4. an optional *next step* when the current result is not yet science-grade.

The catalog and naming are aligned with `AGENT.PIC.TEST.PROBLEM.CATALOG.md`. This note is an internal methods draft; it is not a final science-results paper.

2. TASK LIST: ONE TEST PROBLEM PER TASK

2.1. *Complete Task Index (Current)*

1. **Task 1:** `hydro.hydro_linwave`
2. **Task 2:** `mhd.mhd_linwave`
3. **Task 3:** `mhd.divb_amr`
4. **Task 4:** `particles.pic_entity_deposit_mink`
5. **Task 5:** `particles.pic_entity_deposit_reflect`
6. **Task 6:** `particles.pic_em_vacuum_wave`

7. **Task 7:** `particles.pic_langmuir_frequency_proxy`
8. **Task 8:** `particles.pic_two_stream_growth_proxy`
9. **Task 9:** `particles.pic_weibel_growth_proxy`
10. **Task 10:** `particles.pic_no_mhd_boris`
11. **Task 11:** `particles.pic_boris_midpoint_eb`
12. **Task 12:** `particles.pic_bell_growth_proxy`
13. **Task 13:** `particles.pic_multispecies_backreaction_osc`
14. **Task 14:** `particles.pic_crsi_deltaf_proxy`
15. **Task 15:** `particles.pic_crpai_polarization_proxy`
16. **Task 16:** `particles.pic_expanding_box_anisotropy_proxy`
17. **Task 17:** `particles.pic_refinement_boundary_character`
18. **Task 18:** `particles.pic_amr_shock_lb_smoke`
19. **Task 19:** `particles.pic_analysis_utils`
20. **Task 20:** `particles.pic_deposit_conservation`
21. **Task 21:** `particles.pic_decomp_invariance`
22. **Task 22:** `particles.pic_mhd_passive_mode`
23. **Task 23:** `particles.pic_mhd_current_coupling`
24. **Task 24:** `particles.pic_mhd_coupling_decomp`
25. **Task 25:** `particles.pic_mhd_coupling_multilevel`
26. **Task 26:** `particles.pic_mhd_coupling_nonperiodic`
27. **Task 27:** `particles.pic_mhd_restart_fidelity`
28. **Task 28:** `particles.pic_restart_safety_guards`

3. BASELINE HYDRO AND MHD TASKS (TASKS 1–3)

These three tasks are deliberately conservative. If they drift, every later particle claim becomes ambiguous.

TASK 1: HYDRO_LINWAVE

Setup: script `hydro_linwave.py`; input deck `linear_wave_hydro.athinput`. *Logic.* The test isolates pure hydrodynamic wave transport and asks whether L1 errors fall cleanly with resolution across hydro solver options. *Result.* The gate remains a stable convergence check in the suite and provides order-of-accuracy trends for the hydro branches. *Implication.* Hydrodynamic transport remains a trustworthy base layer for later PIC coupling work. *Next step.* Add a compact publication panel for this task so the baseline is visible in the same figure bundle as the PIC results.

TASK 2: MHD_LINWAVE

Setup: script `mhd_linwave.py`; input deck `linear_wave_mhd.athinput`. *Logic.* The test probes fast, Alfvén, slow, and entropy branches to verify that MHD wave families converge without branch-specific pathology. *Result.* Current gating continues to report clean L1 behavior across the branch set. *Implication.* The core MHD transport operator remains credible, so instability and coupling behavior in later tests is less likely to be contaminated by baseline wave errors. *Next step.* Expose a small multi-branch summary plot in the publication bundle to make this evidence directly visible.

TASK 3: DIVB_AMR

Setup: script `divb_amr.py`; input deck `ffc_divb.athinput`. *Logic.* This task stress-tests constrained transport under AMR by enforcing a strict divergence-cleanliness threshold. *Result.* The gate enforces $\max(|\nabla \cdot \mathbf{B}|) < 10^{-10}$ on AMR outputs and remains the strongest scalar health check for magnetic consistency with refinement. *Implication.* When later coupled tests show structure changes, those changes are less likely to be numerical monopoles in disguise. *Next step.* Add a time-history diagnostic of $\nabla \cdot \mathbf{B}$ extrema to show whether cleanliness is merely bounded or actively damped.

4. ENTITY-MIRRORING PIC CORE TASKS (TASKS 4–9)

This block asks a simple question: does AthenaK reproduce the core deposition, wave, and instability signatures expected from the Entity-style reference logic?

TASK 4: PIC_ENTITY_DEPOSIT_MINK

Setup: script `pic_entity_deposit_mink.py`; input deck `pic_entity_deposit_mink.athinput`. *Logic.* The test checks charge, current, and particle-support parity for a Minkowski-style deposition stencil. *Result.* Figure 1 (left) shows an identical support map across ranks, with zero reported support variance and zero $|\text{npN} - \text{np1}|$ differences at the displayed precision. *Implication.* The discrete deposition geometry is decomposition-invariant in this configuration, which is a precondition for credible MPI scaling studies. *Next step.* Expand this task with an off-grid phase sweep so parity is tested away from highly symmetric particle placements.

TASK 5: PIC_ENTITY_DEPOSIT_REFLECT

Setup: script `pic_entity_deposit_reflect.py`; input deck `pic_entity_deposit_reflect.athinput`. *Logic.* This variant probes whether reflecting boundaries preserve the same deposition integrity as periodic or free-space cases. *Result.* Figure 1 (right) shows the expected reflected support pattern with the same decomposition agreement seen in Task 4. *Implication.* Boundary treatment is not introducing rank-dependent artifacts in the measured support stencil. *Next step.* Add a moving-particle reflection sequence to check parity through boundary crossings, not only at a single snapshot.

TASK 6: PIC_EM_VACUUM_WAVE

Setup: script `pic_em_vacuum_wave.py`; input deck `pic_em_vacuum_wave.athinput`. *Logic.* The task checks whether vacuum EM propagation remains accurate when evolved through the PIC field machinery. *Result.* Figure 2 shows monotonic error reduction across the tested resolutions and a near-perfect serial/MPI overlay. *Implication.* The field update and decomposition pipeline preserve both consistency and convergence in the linear vacuum limit. *Next step.* Increase the resolution ladder by one or two points to make formal order fits more stable.

TASK 7: PIC_LANGMUIR_FREQUENCY_PROXY

Setup: script `pic_langmuir_frequency_proxy.py`; input deck `pic_langmuir_frequency_proxy.athinput`. *Logic.* Rather than merely checking sinusoidal shape, this task checks whether the current history recovers the intended plasma frequency. *Result.* The fitted frequencies in the current panel are close to the expected proxy value and serial/MPI traces are indistinguishable over most of the window. *Implication.* The pusher+deposition path reproduces the intended oscillation timescale in this controlled regime. *Next step.*

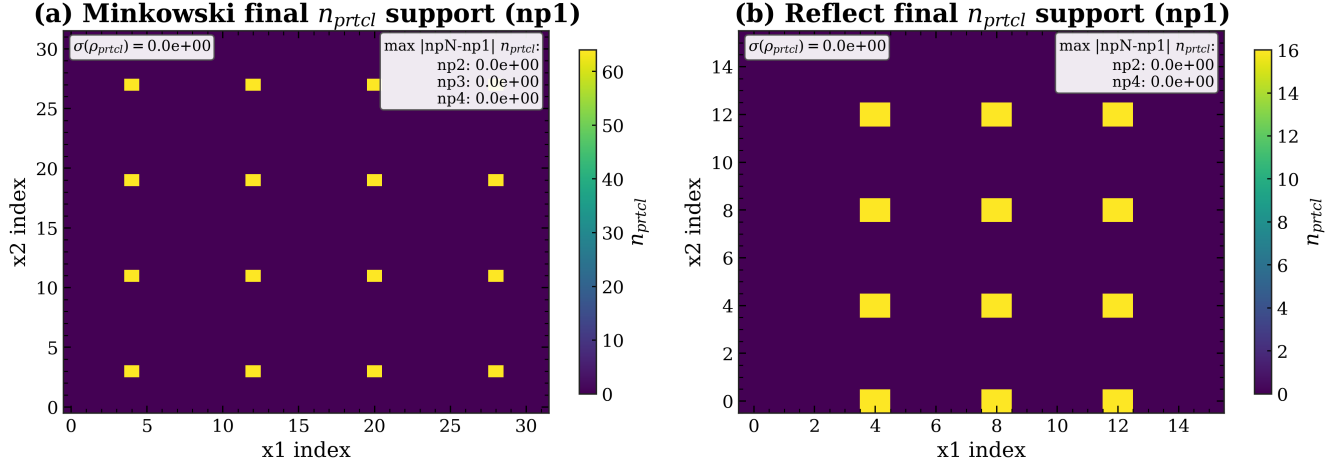


Figure 1. Entity-mirroring deposit support maps for Minkowski and reflecting configurations (Task 4 and Task 5). Bright cells mark the particle-support stencil. Annotations report support variance and decomposition differences; in this generated set those differences are zero at the plotted precision.

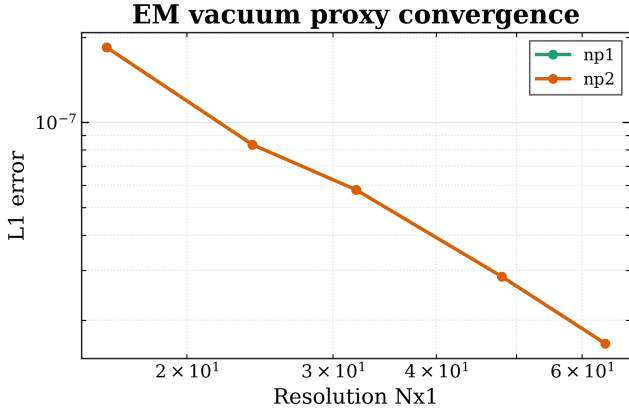


Figure 2. EM-vacuum convergence (Task 6) with serial and MPI overlays. The two curves are visually coincident while L1 error decreases with increasing resolution, indicating decomposition invariance and expected convergence behavior.

Reduce the startup transient (large first-step residual) with a ramped initialization so frequency fits rely on cleaner early-time data.

TASK 8: PIC_TWO_STREAM_GROWTH_PROXY

Setup: script `pic_two_stream_growth_proxy.py`; input deck `pic_two_stream_growth_proxy.athinput`. *Logic.* The task targets unstable electrostatic mode amplification with decomposition-robust behavior. *Result.* Figure 4 (left) shows strong early growth from low amplitude into a nonlinear envelope, with rank-consistent onset but modest late-window divergence. *Implication.* The instability trigger is present and robust, but the sampled window mixes linear and nonlinear phases, so simple single-slope fits are not fully diagnostic. *Next step.* Retune this proxy to hold a longer clean linear

window before saturation for publication-grade growth rate extraction.

TASK 9: PIC_WEIBEL_GROWTH_PROXY

Setup: script `pic_weibel_growth_proxy.py`; input deck `pic_weibel_growth_proxy.athinput`. *Logic.* The task probes magnetic mode amplification in a Weibel-like proxy configuration. *Result.* Figure 4 (right) shows clear growth and rank-tracked trend agreement, again with nonlinear shaping over the longer interval. *Implication.* The solver captures the expected instability pathway, but the present setup is a credibility proxy rather than a final dispersion-relation measurement. *Next step.* Add short-window linear-fit reporting with confidence intervals to separate onset physics from saturation morphology.

5. ATHENA++-INSPIRED MHD-PIC BENCHMARK TASKS (TASKS 10–18)

This section moves from core PIC mechanics into CR-coupled behavior inspired by the Athena++ MHD-PIC benchmark logic. The central question is no longer “does it run,” but “does it run in the right physical direction, with transparent limits?”

TASK 10: PIC_NO_MHD_BORIS

Setup: script `pic_no_mhd_boris.py`; input deck `pic_no_mhd_boris.athinput`. *Logic.* This is the clean single-particle orbit baseline that isolates pusher behavior from MHD coupling effects. *Result.* The gate remains a stable invariant/safety check and is used as a reference for later coupled pusher diagnostics. *Implication.* When coupled runs show drift or phase error, Task 10 helps separate pusher defects from coupling side effects.

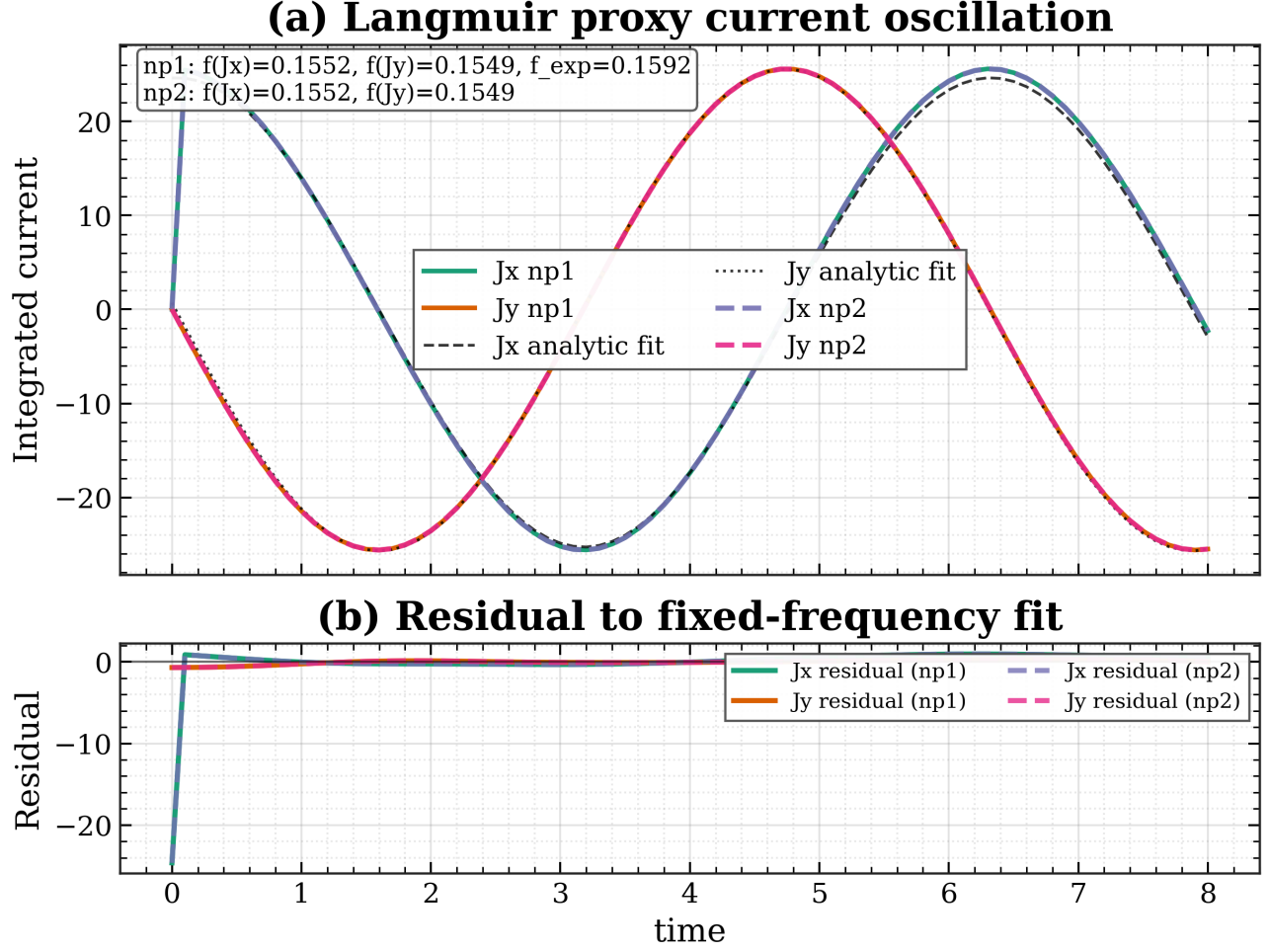


Figure 3. Langmuir-current oscillation and residuals (Task 7). Top panel: integrated J_x and J_y with fixed-frequency analytic fits. Bottom panel: residual to that fit. The dominant recovered frequencies are close to the proxy target and remain aligned between serial and MPI runs.

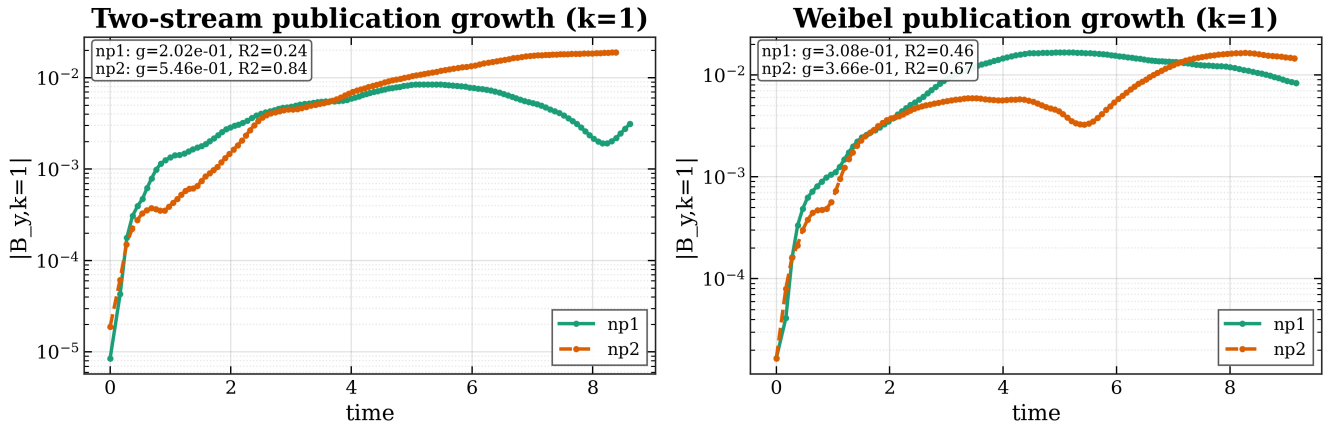


Figure 4. Two-stream and Weibel publication-growth proxies (Tasks 8 and 9). Mode amplitudes are shown on log scales with rank overlays. Both tasks display early growth and later nonlinear shaping, with rank consistency in trend but not perfect overlap in every phase of the trajectory.

Next step. Expose one compact orbit-and-invariant figure in the main bundle so this baseline is visually auditable.

TASK 11: PIC.BORIS_MIDPOINT_EB

Setup: script `pic_boris_midpoint_eb.py`; input deck `pic_boris_midpoint_eb.athinput`. *Logic.* The task verifies the midpoint E+B staging used in the modernized pusher path and checks frozen-in consistency under staged updates. *Result.* Current gating confirms expected momentum/energy-exchange behavior for this controlled setup. *Implication.* The midpoint formulation is integrated coherently with the operator-splitting sequence used by coupled updates. *Next step.* Add an explicit error-versus-step-size curve to quantify pusher convergence rate in this mode.

Bell growth: coupled vs uncoupled (k=1 transverse mode)

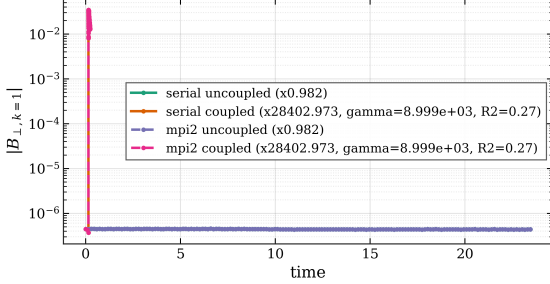


Figure 5. Bell growth proxy (Task 12), shown as $|B_{\perp, k=1}|$ for coupled and uncoupled runs in serial and MPI layouts. Coupled traces rise above uncoupled baseline with strong rank agreement, but the present trajectory is dominated by startup impulse and nonlinear reshaping rather than a long, clean linear-growth segment.

TASK 12: PIC.BELL_GROWTH_PROXY

Setup: script `pic_bell_growth_proxy.py`; input decks `pic_bell_growth_proxy.athinput` and `pic_bell_growth_publication.athinput`. *Logic.* The test asks whether activating CR-MHD coupling produces transverse magnetic amplification while the uncoupled baseline remains quiet. *Result.* Figure 5 confirms this activation cleanly and shows serial/MPI consistency, but the measured curve is not yet a textbook sustained linear-growth regime. *Implication.* Coupling-path correctness is credible; quantitative Bell growth-rate science claims remain premature with the current startup-dominated window. *Next step.* Retune the initial perturbation, CR loading, and fit window to isolate a longer linear segment before nonlinear rollover.

TASK 13: PIC.MULTISPECIES_BACKREACTION_OSCILLATION

Setup: script `pic_multispecies_backreaction_oscillation.py`; input decks `pic_multispecies_osc_uniform.athinput`,

`pic_multispecies_osc_smr.athinput`, and `pic_multispecies_osc_amr_proxy.athinput` plus publication variants. *Logic.* The task checks backreaction-mediated oscillation behavior across mesh hierarchies and tracks associated energy drift. *Result.* Figure 6 shows phase-coherent early oscillation with mesh-dependent damping and drift plateaus of order 10^{-2} to a few 10^{-2} . *Implication.* The method is stable and comparable across mesh strategies at proxy level, but long-window energy conservation is not yet tight enough for precision claims. *Next step.* Tighten drift via timestep and deposition tuning, then repeat the uniform/SMR/AMR comparison with longer windows and uncertainty bars.

TASK 14: PIC.CRSI_DELTAF_PROXY

Setup: script `pic_crsi_deltaf_proxy.py`; input decks `pic_crsi_deltaf_proxy.athinput` and `pic_crsi_deltaf_publication.athinput`. *Logic.* The test uses δf weighting to reduce noise and examine polarization-branch behavior in a CR streaming instability proxy. *Result.* Figure 7 shows strong rank agreement in the current tuned setup and no catastrophic branch misalignment across np1/np2/np4. *Implication.* Decomposition sensitivity is reduced relative to earlier proxy runs, which strengthens confidence in the analysis path and weighting implementation. *Next step.* Establish a parameter window where branch hierarchy and growth are both stable over longer times so this proxy can support stronger physics interpretation.

TASK 15: PIC.CRPai_POLARIZATION_PROXY

Setup: script `pic_crpai_polarization_proxy.py`; input decks `pic_crpai_prolate_proxy.athinput` and `pic_crpai_oblate_proxy.athinput`, plus publication variants. *Logic.* This task probes how anisotropy geometry shifts polarization response and branch dominance. *Result.* Figure 8 shows that prolate and oblate cases produce distinct branch behavior while preserving strong rank overlap in the tuned window. *Implication.* The implementation captures geometry-dependent polarization response without obvious decomposition artifacts. *Next step.* Extend the runtime and scan anisotropy parameters so branch crossing and late-time branch competition can be measured, not only observed.

TASK 16: PIC.EXPANDING_BOX_ANISOTROPY_PROXY

Setup: script `pic_expanding_box_anisotropy_proxy.py`; input decks `pic_expanding_box_proxy.athinput` and `pic_compressing_box_proxy.athinput`. *Logic.* The test checks whether kinetic anisotropy evolves in opposite directions under expansion and compression. *Result.* Figure 9 reports a negative expansion slope and

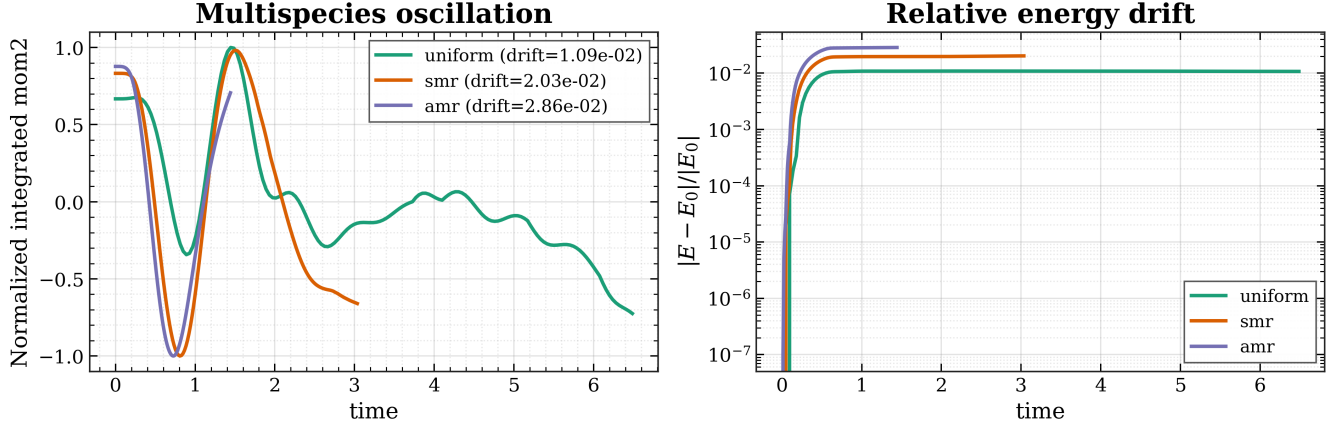


Figure 6. Multispecies backreaction oscillation diagnostics (Task 13). Left: normalized integrated momentum for uniform, SMR, and AMR runs. Right: relative energy drift. Oscillation phase behavior is qualitatively consistent, while drift plateaus differ by mesh strategy.

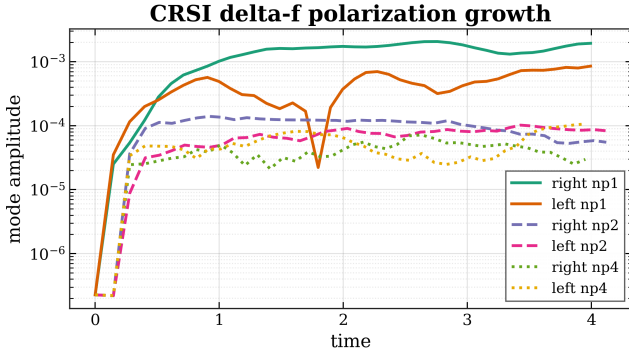


Figure 7. CRSI δf polarization growth proxy (Task 14). Right and left branches are overlaid across np1/np2/np4 decompositions. The current tuned run shows close rank collapse and reduced branch asymmetry over the sampled window, though this remains a proxy-level check.

positive compression slope with small fit uncertainties. *Implication.* The sign-level physical response is robust in this proxy regime, which is exactly what this gate is meant to establish. *Next step.* Add a multi-rate scan in $\dot{a}$ to connect these proxy slopes to scaling trends.

TASK 17: PIC_REFINEMENT_BOUNDARY_CHARACTERIZATION

Setup: script `pic_refinement_boundary_characterization.py`, input deck `pic_refinement_boundary_smr_proxy.athinput`. Comparison deck `pic_refinement_boundary_amr_proxy.athinput`.

Logic. The task measures how deposition smoothness and conservation behave across coarse/fine interfaces. *Result.* The gate tracks coefficient-of-variation, jump metrics, and drift behavior and remains a required integration checkpoint in the suite. *Implication.* Refinement boundaries are being monitored as a first-class risk, which is necessary for any AMR-heavy science campaign. *Next step.* Publish a dedicated boundary

diagnostic figure so this task has the same visual traceability as the instability proxies.

TASK 18: PIC_AMR_SHOCK_LB_SMOKE

Setup: script `pic_amr_shock_lb_smoke.py`; input deck `pic_amr_shock_lb_smoke.athinput`. Publication decks: `pic_amr_shock_lb_publication_local.athinput` and `pic_amr_shock_lb_publication_hpc.athinput`. *Logic.* The test asks whether coupled CR-shock execution survives AMR and load balancing while still showing recognizable shock morphology. *Result.* Figure 10 shows coherent density and magnetic structure with nontrivial current patterns, but no phase-space spectrum in the fallback output. *Implication.* The infrastructure path is healthy; the case is not yet publication-grade shock physics. *Next step.* Run longer, higher-dynamic-range campaigns with particle phase-space and spectral outputs enabled, then build the full multi-panel shock storyline.

6. INTEGRATION AND SAFETY TASKS (TASKS 19–28)

These tasks are the suite’s integrity scaffolding. Most do not produce headline figures, but they protect every figure that does.

TASK 19: PIC_ANALYSIS_UTILS

Setup: script `pic_analysis_utils.py`. *Logic.* Validate the shared fitting and reduction utilities used by publication plotting scripts. *Result.* The task remains a dedicated self-test for analysis kernels. *Implication.* Numerical regressions in fit routines are caught before they silently alter interpretation plots. *Next step.* Add stricter regression fixtures for uncertainty estimates and fit-window sensitivity.

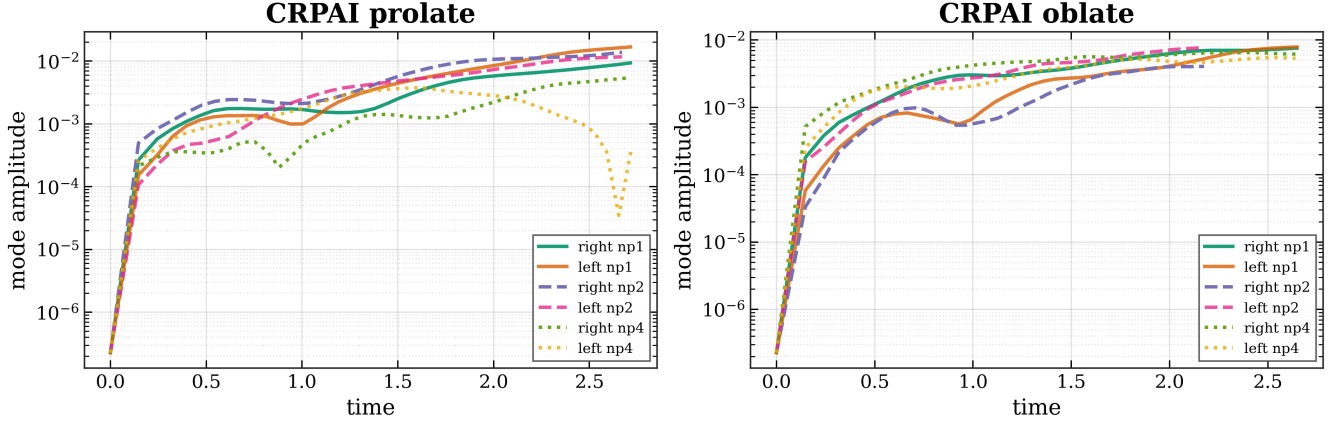


Figure 8. CRPAI polarization proxy (Task 15) for prolate (left) and oblate (right) initial distributions. Right/left branches for np1/np2/np4 are plotted on each panel. Rank collapse is strong in the present tuning; branch ordering changes with anisotropy geometry as expected.

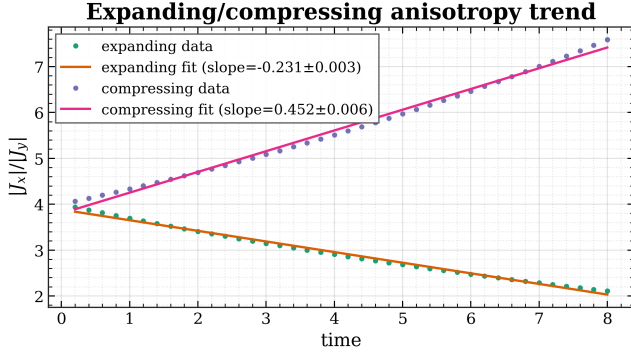


Figure 9. Expanding/compressing-box anisotropy trend (Task 16). Data points are fit by lines with slope uncertainties. Expansion and compression produce opposite-sign slopes with clear separation over the full sampled interval.

TASK 20: PIC_DEPOSIT_CONSERVATION

Setup: script `pic_deposit_conservation.py`; input deck `pic_deposit_conservation.athinput`. *Logic.* Check deposition conservation for charge and moments, including expected negative-path parser cases. *Result.* The task continues to serve as a scalar conservation gate and error-path validator. *Implication.* Core deposition bookkeeping remains under continuous watch, not inferred from higher-level plots. *Next step.* Expose a small conservation-time-history figure for publication transparency.

TASK 21: PIC_DECOMP_INVARIANCE

Setup: script `pic_decomp_invariance.py`; input deck `pic_deposit_conservation.athinput`. *Logic.* The task asks whether integrated deposited moments remain invariant when MPI decomposition changes. *Result.* Current gating checks scalar differences against strict tolerances. *Implication.* Any decomposition-

induced bias is promoted to a hard regression, not treated as acceptable run-to-run noise. *Next step.* Report invariance metrics in a compact table in this manuscript revision cycle.

TASK 22: PIC_MHD_PASSIVE_MODE

Setup: script `pic_mhd_passive_mode.py`; input deck `pic_mhd_passive_mode.athinput`. *Logic.* Verify that passive-mode operation does not accidentally activate coupling pathways. *Result.* The gate checks mode-specific guard behavior and expected passive evolution signatures. *Implication.* Feature boundaries are explicit, reducing accidental cross-contamination between passive and coupled runs. *Next step.* Add direct comparison plots of passive versus coupled channels for clearer operator separation evidence.

TASK 23: PIC_MHD_CURRENT_COUPLING

Setup: script `pic_mhd_current_coupling.py`; input decks `pic_mhd_current_coupling.athinput` and `pic_mhd_current_coupling.default_mode.athinput`. *Logic.* This is the main correctness gate for current coupling, including direct-versus-conversion representation checks and continuity behavior. *Result.* The task continues to enforce parser guards, mode behavior, and representation consistency. *Implication.* The coupling API is constrained by explicit mechanical checks, not only by end-state agreement. *Next step.* Add a dedicated publication panel that tracks direct/conversion ratios through time.

TASK 24: PIC_MHD_COUPLING_DECOMP

Setup: script `pic_mhd_coupling_decomp.py`; input deck `pic_mhd_current_coupling.athinput`. *Logic.* Probe whether coupled quantities retain continuity and expected values under decomposition changes. *Result.*

CR-shock storyline quick-look (smoke-fallback)

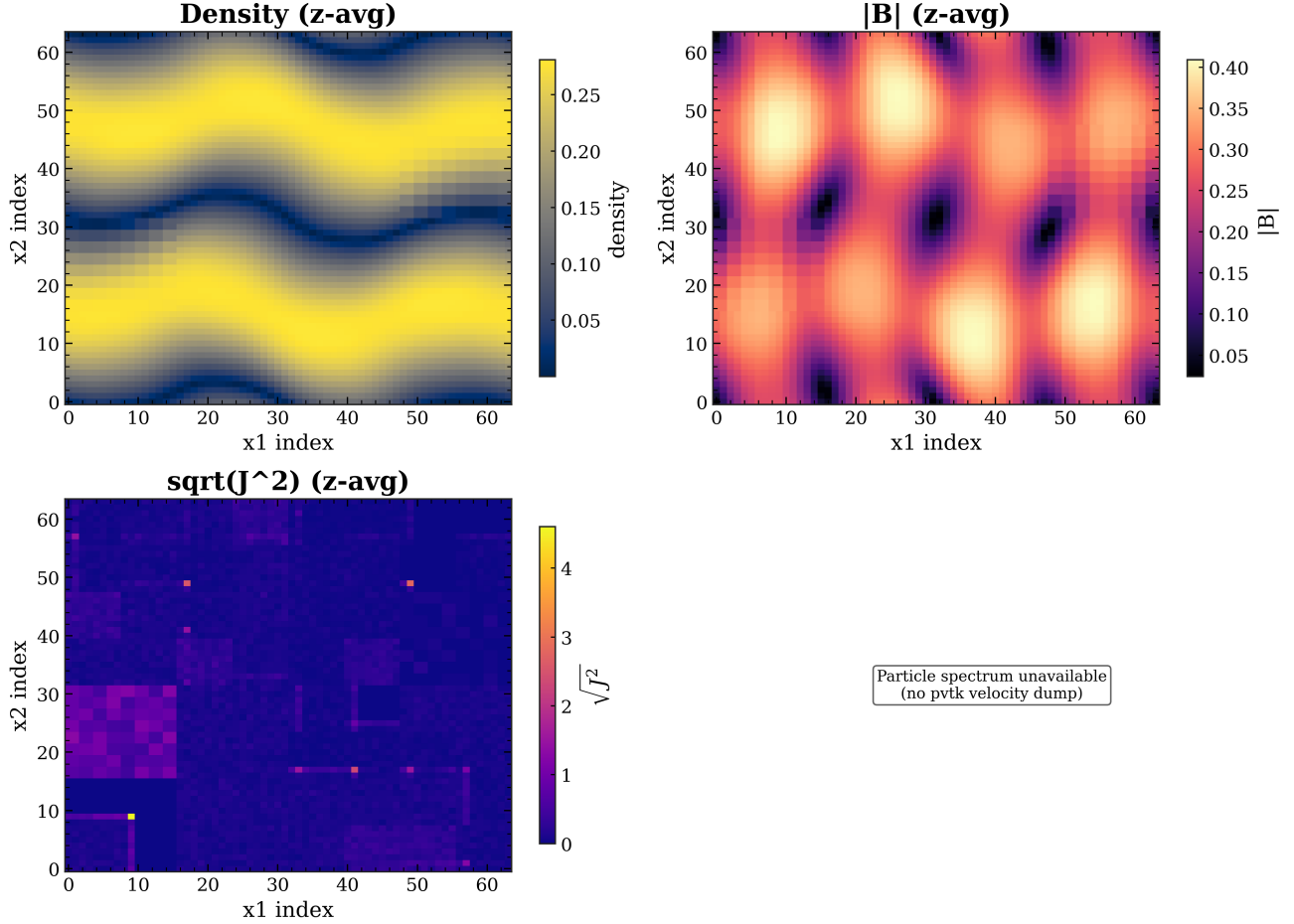


Figure 10. Reduced AMR-shock storyline panel (Task 18, smoke-level fallback). Shown are density, $|B|$, and $\sqrt{J^2}$ maps. The panel demonstrates organized coupled structure under AMR/load-balance execution. A particle spectrum panel is still absent in this fallback product, so the figure is implementation evidence rather than a final science narrative.

Current gate compares coupled diagnostics across rank layouts. *Implication.* MPI decomposition is treated as a numerical coordinate transform, not a hidden model parameter. *Next step.* Promote key decomposition deltas into a persistent trend log so drift can be seen before thresholds are crossed.

TASK 25: PIC_MHD_COUPLING_MULTILEVEL

Setup: script `pic_mhd_coupling_multilevel.py`; input deck `pic_mhd_coupling_multilevel.athinput`. *Logic.* Test coupled behavior across mesh hierarchy changes and multilevel interpolation pathways. *Result.* The task remains the main scalar parity check for multilevel coupling consistency. *Implication.* Mesh hierarchy effects are directly tested, which is critical before AMR-heavy physics campaigns. *Next step.* Add level-by-level coupling budget plots for easier diagnosis of where hierarchy errors begin.

TASK 26: PIC_MHD_COUPLING_NONPERIODIC

Setup: script `pic_mhd_coupling_nonperiodic.py`; input decks `pic_mhd_coupling_nonperiodic.athinput` and `pic_mhd_coupling_nonperiodic_default_mode.athinput`. *Logic.* Check nonperiodic boundary behavior and verify that unsupported combinations fail through intentional guard paths. *Result.* The task enforces both positive-path behavior and negative-path guard correctness. *Implication.* Boundary-condition support is explicit and test-backed, reducing silent misuse in production runs. *Next step.* Add targeted edge-case decks for mixed boundary combinations that are close to supported configurations.

TASK 27: PIC_MHD_RESTART_FIDELITY

Setup: script `pic_mhd_restart_fidelity.py`; input deck `pic_mhd_restart_fidelity.athinput`. *Logic.* Compare full versus segmented trajectories to verify restart equivalence in coupled runs. *Result.* Current

gating checks restart-parity metrics and representation ratios. *Implication.* Restart state transfer is being validated as part of numerical correctness, not deferred to operations practice. *Next step.* Expand reporting on per-rank restart restore messages to close the remaining watch item with clearer diagnostics.

TASK 28: PIC_RESTART_SAFETY_GUARDS

Setup: script `pic_restart_safety_guards.py`; input deck `pic_restart_safety_guards.athinput`. *Logic.* This task verifies that restart misuse and unsafe configurations fail loudly and predictably. *Result.* The gate preserves expected-failure coverage for restart safety paths. *Implication.* Failure behavior is part of the tested contract, which improves operational trust in long coupled campaigns. *Next step.* Add matrix coverage for combined restart+decomposition edge cases to ensure guard behavior remains consistent under MPI variation.

7. CURRENT FIGURE BUNDLE AND REPRODUCIBILITY

The figures in this draft were generated from `tst/.codex/pic_publication_runs/20260208_113729` using `plot_pic_publication_figures.py` and copied into this manuscript package. The bundle currently includes ten publication-facing figures:

1. Entity deposition support maps,
2. EM-vacuum convergence,
3. Langmuir oscillation with residuals,
4. two-stream and Weibel mode traces,
5. Bell growth proxy,
6. multispecies oscillation and drift,
7. CRSI polarization,
8. CRPAI polarization,
9. expanding/compressing anisotropy trend,
10. reduced shock storyline panel.

8. SYNTHESIS: WHAT THE SUITE SAYS NOW

The suite now supports three statements with high confidence.

1. **Core numerics are coherent.** Baseline hydro/MHD checks and Entity-mirroring deposition tasks show stable transport and deposition book-keeping.

2. **Coupling mechanics are active and test-backed.** Coupled proxies (Bell, CRSI, CRPAI, multispecies) show expected directional behavior and strong rank consistency in the tuned windows.
3. **Infrastructure for bigger science runs is present but not yet exhausted.** The AMR shock storyline runs through the full machinery, yet still lacks the long-run and phase-space depth required for final CR-shock science figures.

A careful reading still leaves two practical cautions. First, several instability proxies currently mix startup transients, linear growth, and nonlinear rollover in one window; that is useful for implementation checks but not ideal for extracting robust physical growth rates. Second, the shock case is already informative as a systems test, but it is still a smoke-level implementation product until richer CR diagnostics are enabled and scaled.

9. IMMEDIATE NEXT IMPROVEMENTS

The shortest path to publication-grade completeness is:

1. **Instability-window redesign:** retune Bell/CRSI/CRPAI windows for cleaner linear-growth intervals and stronger fit diagnostics.
2. **Shock-depth upgrade:** run the publication shock decks long enough to recover downstream spectra and $f(x,p)$ structure, then build the full multi-panel storyline.
3. **Baseline-visibility polish:** add compact baseline hydro/MHD and boundary-characterization figures so every major gate family has a visual anchor in the atlas.

This draft is therefore both record and trajectory: it captures exactly what is already working, and it marks where targeted refinement turns method confidence into publication-level physics evidence.

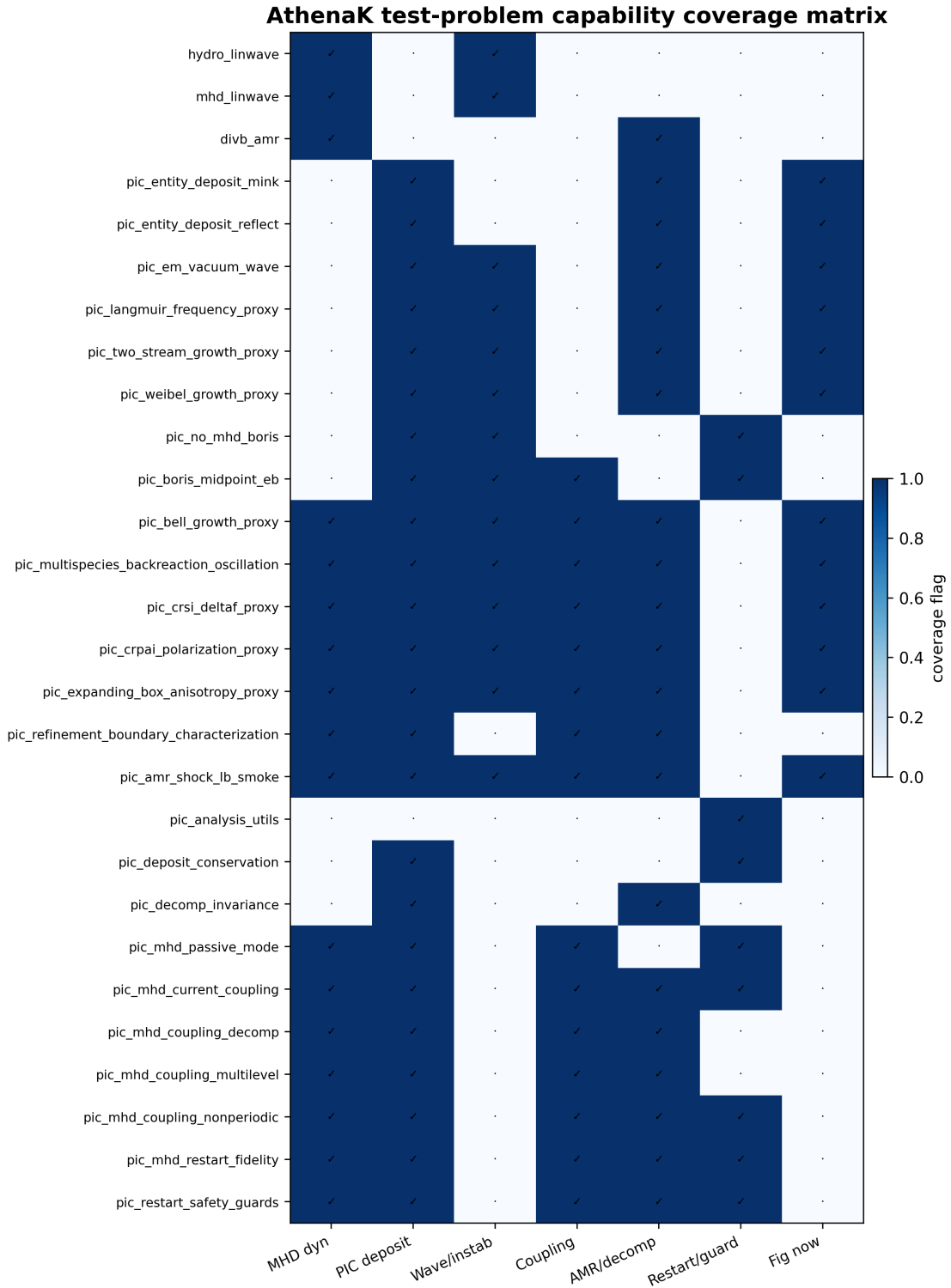


Figure 11. Capability coverage matrix for all 28 tasks. Rows map one-to-one to Section 2 tasks; columns summarize capability classes: MHD dynamics, PIC deposition, wave/instability behavior, coupling fidelity, AMR/decomposition, restart/guard safety, and figure availability. The matrix is a navigation map, not a substitute for physics diagnostics.